

FPGAs NOT JUST LOGIC GATES ANY MORE!

While the debate on ASIC versus FPGA continues, today's FPGAs with integrated functional blocks like high-speed transceivers, DSP blocks, hard IPs like PCIe express and Ethernet MAC, and DDR3 memory controllers have surpassed ASICs in many embedded applications. These have now become designers' choice in communication, defence, medical, consumer electronics and many more devices

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Earlier, if we were to summarise the ASIC versus FPGA battle for cost and performance, ASICs had a greater penetration because there was demand for high-volume applications in which application-specific integrated circuits (ASICs) provided a cost-effective solution and field-programmable gate arrays (FPGAs) were a costlier affair. However, now we see a reversal in trend. Experts say that the cost of entry for ASIC design is skyrocketing, which has made FPGAs a natural choice. Apart from the cost, the reduced time-to-market and design flexibility have made FPGAs relevant for more and more embedded applications.

Traditionally, ASICs were preferred in high-volume applications because of the overall low cost of production. "Today, the demand for ultra-high-volume products is reducing and so is the demand for ASICs," shares Neeraj Varma, country manager, Xilinx India. "Last year use of ASICs declined by 21 per cent for the simple reason of exorbitant cost associated with them," says Varma. Market segments like communication (routers, switches, back-haul equipment, and wireless pico and femto cells) and high-end consumer

electronics (flat-panel plasma TVs) centre around FPGAs. Medical, industrial, defence, security and surveillance related applications are also designed using FPGAs.

"Use of FPGAs offers great flexibility, scalability/reprogrammability and almost zero non-recurring engineering (NRE) cost," explains Apurva Prabhakar, senior design engineer,

off-the-shelf. With an extensive library of intellectual property (IP) available for use, you can change design in hours."

Prashant Aggarwal, vice president-technology, VirtualWire Technologies, comments, "The design and development cycle for an ASIC design is 2-3 years, while with FPGA it's few months to about a year." This helps in serving the constantly changing requirement of the market.

FPGAs aren't just the logic gates

The first FPGA from XC2064 had 64 configurable logic blocks and two 3-input look-up tables (LUTs). Today, we have FPGAs with millions of logic cells and many integrated blocks in a very tiny area. In the electronics industry, something that is programmable and switchable was always available in terms of discrete logic components, programmable logic devices (PLDs), etc.

Varma comments, "The whole idea was to replace or integrate the logic components (what we refer to as glue logic) into a single component." As the silicon technology developed, the number of logic cells, capabilities and the speed of FPGAs increased.

When FPGAs were introduced, these were blank hardware chips. "With advancements in technology, FPGAs included more and more

embedded group, KritiKal Solutions. She adds, "For designing with FPGAs, you have easy-to-use and less complex design tools. For simple designs, a single integrated tool for simulation, synthesis, place-and-route and binary file generation can be used."

Adding to the benefits, Barry Tso, regional manager, Avnet Electronics Marketing Asia, says, "FPGA is a standard product and easily available

